OS hardening: patching and rescan validation

Securing Systems and Data | Full Sail University

Academic coursework. The company and its server are fictional and part of the course scenario. All scanning was performed against isolated lab systems.

Includes the lab write-up and the follow-up Nessus scan report (0 Critical, 0 High, 1 Medium after patching).

Redacted for publication: IP addresses, instructor name. Black boxes mark removed content.

Lab 3.1 – Lab 3 – OS Hardening

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WRITE UP

1. After updating the Hammercorp server and rescanning with Nessus

a. How many vulnerabilities were left?

After patching the Hammer Corp Server and conducting a second scan we only found 1 Medium vulnerability remaining. This a huge difference from the initial scan of 46 Critical and 48 high vulnerabilities.

b. How does this make you feel about the importance of regular patching?

Honestly, it showered me the importance of regular patching and paying attention to these very necessary updates that OS need. It really is a non-negotiable step when hardening a system.

2. Provide a sentence or two answering the following questions:

a. Was the HammerCorp Corporate server "secure" based on the scans? Why or why not?

No, there is a difference between a "secure" and a better security posture. Plus, a system with any vulnerability, which in this case we found 1 remaining medium vulnerability, makes this system not "secure". While all the high and critical vulnerabilities are gone this last vulnerability can still be exploited.

b. Based on the above question what do you think should be done before placing a system in production?

I'd say before moving a system to production phase you need check mark several things but primarily:
-Verify that the system is fully patched with all the OS, and software updates.
-Conduct a vulnerability scan with whatever tool of your liking or the one that the organization uses.
-Lastly conduct a QA to establish security protocols and a QC for the final inspection of test to make sure the product is meeting the necessary security standards.

tenable Nessus

Report generated by Tenable Nessus™

HammerCorpServer Scan

Sun, 19 Oct 2025 14:44:51 EDT

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Vulnerabilities by Host

- [redacted]

Vulnerabilities by Host

Collapse All | Expand All

[redacted]

0	0	1	0	24
CRITICAL	HIGH	MEDIUM	LOW	INFO

Severity	CVSS v3.0	VPR Score	EPSS Score	Plugin	Name
MEDIUM	5.3	-	-	57608	SMB Signing not required
INFO	N/A	-	-	45590	Common Platform Enumeration (CPE)
INFO	N/A	-	-	10736	DCE Services Enumeration
INFO	N/A	-	-	54615	Device Type
INFO	N/A	-	-	35716	Ethernet Card Manufacturer Detection
INFO	N/A	-	-	86420	Ethernet MAC Addresses
INFO	N/A	-	-	10092	FTP Server Detection
INFO	N/A	-	-	43111	HTTP Methods Allowed (per directory)
INFO	N/A	-	-	10107	HTTP Server Type and Version
INFO	N/A	-	-	24260	HyperText Transfer Protocol (HTTP) Information
INFO	N/A	-	-	10785	Microsoft Windows SMB NativeLanManager Remote System Information Disclosure
INFO	N/A	-	-	11011	Microsoft Windows SMB Service Detection
INFO	N/A	-	-	100871	Microsoft Windows SMB Versions Supported (remote check)
INFO	N/A	-	-	106716	Microsoft Windows SMB2 and SMB3 Dialects Supported (remote check)

Nessus Essentials

INFO	N/A	-	-	11219	Nessus SYN scanner
INFO	N/A	-	-	19506	Nessus Scan Information
INFO	N/A	-	-	209654	OS Fingerprints Detected
INFO	N/A	-	-	11936	OS Identification
INFO	N/A	-	-	117886	OS Security Patch Assessment Not Available
INFO	N/A	-	-	22964	Service Detection
INFO	N/A	-	-	110723	Target Credential Status by Authentication Protocol - No Credentials Provided
INFO	N/A	-	-	10287	Traceroute Information
INFO	N/A	-	-	33139	WS-Management Server Detection
INFO	N/A	-	-	11422	Web Server Unconfigured - Default Install Page Present
INFO	N/A	-	-	10150	Windows NetBIOS / SMB Remote Host Information Disclosure

* indicates the v3.0 score was not available; the v2.0 score is shown

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